

Web Programming

Computing and Mathematics

May, 2026

Web Programming (A13843)

Short Title:	Web Programming	
Faculty	Science and Computing	
Department	Computing and Mathematics	
Cluster	Software and Web Development	
Author	RBIRNEY	
Credits	5	Level: Advanced

Description of Module / Aims

This module introduces the student to the principles of web programming. The student will use a responsive framework to develop a web front-end. Students will also be introduced to client-side and server-side scripting techniques and will gain competency in developing full stack web applications that incorporate database access.

Programmes

COMP-0623	BSc in Applied Computing (WD_KCOMP_D)	3 / 6 / M
COMP-0623	Higher Diploma in Science in Business Systems Analysis (WD_KBUSY_G)	4 / 2 / M

Indicative Content

- Responsive web front-end frameworks
- Client-side scripting: interactivity; client side data processing
- Server-side programming: database connectivity; CRUD (Create, Read, Update, Delete) functions
- Publishing web pages using FTP (File Transfer Protocol)
- Website deployment
- Website security

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Develop a web front-end using a responsive framework.
2. Construct and deploy a web application based on the client/server application model.
3. Develop a web application that utilises client-side scripting techniques.
4. Integrate database access in a web application using server side scripting techniques.

Learning and Teaching Methods

- The lectures will introduce the theory content to the student.
- The student will be encouraged to participate in class discussions and ask questions to support their learning process.
- The practical classes facilitate the student in implementing the theory learned in the lectures, and allow them to develop practical skills in web development.
- Supplementary material will be accessible online.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Project	100%	1,2,3,4

Assessment Criteria

<40%: Unable to interpret and describe key concepts that relate to Web Programming.

40%-49%: Be able to interpret and describe key concepts that relate to Web Programming.

50%-59%: Ability to discuss key concepts that relate to Web Programming and ability to discover and integrate related knowledge in other knowledge domains.

60%-69%: Be able to solve problems that relate to Web Programming by experimenting with the appropriate skills and tools.

70%-100%: All the above to an excellent level. Be able to analyse Web Programming solutions to a high standard for a range of both complex and unforeseen problems through the use and modification of appropriate skills and tools.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	12
Practical	24	12
Independent Learning	87	111

Supplementary Material

- Brinzarea, B., C. Darie and A. Hendrix. *_AJAX and PHP: Building Modern Web Applications_*. 2nd ed. Birmingham: Packt Publishing, 2010.
- Nixon, R. *_Learning PHP, MySQL & JavaScript: With jQuery, CSS & HTML5_*. 4th ed. Sebastopol, CA: O' Reilly Media, 2014.

Requested Resources

- Room Type: Computer Lab